Assignment #6 Solutions
Riemannian and Complex Geometry

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1 Hermitian Metric

Exercise 6.1

Problem: Let (E, h) be a Hermitian vector bundle over a complex manifold X. A connection ∇ is a metric connection if and only if

$$d\{s,t\} = \{\nabla s, t\} + (-1)^p\{s, \nabla t\},$$

where $s \in C^\infty(X, \Omega_X^p \otimes E)$ and $t \in C^\infty(X, \Omega_X^q \otimes E)$.

Proof. We need to prove both directions of the equivalence.

($\Rightarrow$) Assume ∇ is a metric connection.

By definition, a connection ∇ on E is metric with respect to the Hermitian metric h if for any vector field V on X and sections $s, t \in C^\infty(X, E)$:

$$V \cdot h(s,t) = h(\nabla_V s, t) + h(s, \nabla_V t)$$

Let $s \in C^\infty(X, \Omega_X^p \otimes E)$ and $t \in C^\infty(X, \Omega_X^q \otimes E)$. The pairing $\{s,t\}$ is a $(p+q)$-form with values in $\mathbb{C}$ defined by:

$$\{s,t\}(v_1, \ldots, v_{p+q}) = h(s(v_1, \ldots, v_p), t(v_{p+1}, \ldots, v_{p+q}))$$

Taking the exterior derivative d of this pairing:

$$\begin{aligned} d\{s,t\}(v_0, \ldots, v_{p+q}) &= \sum_{i=0}^{p+q} (-1)^i v_i \cdot \{s,t\}(v_0, \ldots, \hat{v}_i, \ldots, v_{p+q}) \\ &\quad + \sum_{i<j} (-1)^{i+j} \{s,t\}([v_i, v_j], v_0, \ldots, \hat{v}_i, \ldots, \hat{v}_j, \ldots, v_{p+q}) \end{aligned}$$

Using the metric property of ∇:

$$\begin{aligned} v_i \cdot \{s,t\} &= v_i \cdot h(s,t) \\ &= h(\nabla_{v_i} s, t) + h(s, \nabla_{v_i} t) \\ &= \{\nabla_{v_i} s, t\} + \{s, \nabla_{v_i} t\} \end{aligned}$$

For $s \in \Omega^p \otimes E$, we have:

$$(\nabla s)(v_0, \ldots, v_p) = \sum_{i=0}^{p} (-1)^i \nabla_{v_i} s(v_0, \ldots, \hat{v}_i, \ldots, v_p) + \text{curvature terms}$$

Combining these calculations and accounting for the sign $(-1)^p$ that arises from moving s past p vector fields:

$$d\{s,t\} = \{\nabla s, t\} + (-1)^p\{s, \nabla t\}$$

($\Leftarrow$) Assume $d\{s,t\} = \{\nabla s, t\} + (-1)^p\{s, \nabla t\}$ for all s, t.

Taking $p = q = 0$, so $s, t \in C^\infty(X, E)$ are just sections:

$$d\{s,t\} = \{\nabla s, t\} + \{s, \nabla t\}$$

For a vector field V:

$$V \cdot \{s,t\} = h(\nabla_V s, t) + h(s, \nabla_V t)$$

This is precisely the definition of a metric connection. Therefore, ∇ is metric. □

Exercise 6.2

Problem: Let (E, h) be a Hermitian vector bundle equipped with connection ∇^E. Then ∇^E is a metric connection if and only if $\nabla^{E^* \otimes E} h = 0$.

Proof. Recall that the Hermitian metric h can be viewed as a section of $E^* \otimes \overline{E}^*$ or equivalently $E^* \otimes E^*$ after complexification.

($\Rightarrow$) Assume ∇^E is metric.

Let s, t be local sections of E. The metric h defines a pairing:

$$h : E \times E \to \mathbb{C}$$

We need to show that $\nabla^{E^* \otimes E} h = 0$. The induced connection on $E^* \otimes E$ acts on h by:

$$(\nabla^{E^* \otimes E} h)(s, t) = \nabla(h(s, t)) - h(\nabla^E s, t) - h(s, \nabla^E t)$$

Since ∇^E is metric, by definition for any vector field V:

$$V(h(s, t)) = h(\nabla^E_V s, t) + h(s, \nabla^E_V t)$$

This means:

$$\nabla(h(s, t)) = h(\nabla^E s, t) + h(s, \nabla^E t)$$

Therefore:

$$(\nabla^{E^* \otimes E} h)(s, t) = 0$$

Hence $\nabla^{E^* \otimes E} h = 0$.

($\Leftarrow$) Assume $\nabla^{E^* \otimes E} h = 0$.

This means for any sections s, t of E:

$$\nabla(h(s, t)) = h(\nabla^E s, t) + h(s, \nabla^E t)$$

For any vector field V:

$$V(h(s, t)) = h(\nabla^E_V s, t) + h(s, \nabla^E_V t)$$

This is precisely the condition for ∇^E to be a metric connection. □

Exercise 6.3

Problem: Let $\mathbb{CP}^n = \bigcup_{i=0}^n U_i$ be the canonical open covering, where $U_i = \{(z^0 : \cdots : z^n) \mid z^i \neq 0\}$. Show that there is a Hermitian metric h on the holomorphic tangent bundle of $\mathbb{CP}^n$, called the Fubini-Study metric, such that:

$$\omega|_{U_i} = \frac{\sqrt{-1}}{2} \partial \bar{\partial} \log \frac{\sum_{l=0}^n |z^l|^2}{|z^i|^2}$$

where $\omega = h_{ij} dz^i \wedge d\bar{z}^j$.

Proof. **Step 1: Define the Fubini-Study metric.**

On U_i, we use local coordinates $w^j = z^j / z^i$ for $j \neq i$. The Fubini-Study Kähler potential is:

$$\phi_i = \log \left(\sum_{l=0}^n |z^l|^2 \right) - \log |z^i|^2 = \log \left(1 + \sum_{j \neq i} |w^j|^2 \right)$$

The Fubini-Study form is:

$$\omega = \frac{\sqrt{-1}}{2}\partial\bar{\partial}\phi_i$$

Step 2: Compute $\partial\bar{\partial}\phi_i$ in detail.
Let $\rho = 1 + \sum_{j\neq i} |w^j|^2 = 1 + \sum_{j\neq i} w^j \bar{w}^j$. Then $\phi_i = \log\rho$.
Substep 2.1: Compute $\partial\rho$.

$$\begin{aligned}
\partial\rho &= \partial\left(1 + \sum_{j\neq i} w^j\bar{w}^j\right) \\
&= \sum_{j\neq i} \partial(w^j\bar{w}^j) \\
&= \sum_{j\neq i} (\partial w^j)\bar{w}^j \quad (\text{since } \partial\bar{w}^j = 0) \\
&= \sum_{j\neq i} \bar{w}^j dw^j
\end{aligned}$$

Substep 2.2: Compute $\bar{\partial}\rho$.

$$\begin{aligned}
\bar{\partial}\rho &= \bar{\partial}\left(1 + \sum_{j\neq i} w^j\bar{w}^j\right) \\
&= \sum_{j\neq i} w^j(\bar{\partial}\bar{w}^j) \quad (\text{since } \bar{\partial}w^j = 0) \\
&= \sum_{j\neq i} w^j d\bar{w}^j
\end{aligned}$$

Substep 2.3: Compute $\partial\phi_i = \partial(\log\rho)$.

$$\begin{aligned}
\partial\phi_i &= \partial(\log\rho) \\
&= \frac{1}{\rho}\partial\rho \\
&= \frac{1}{\rho}\sum_{j\neq i} \bar{w}^j dw^j \\
&= \sum_{j\neq i} \frac{\bar{w}^j}{1 + \sum_{k\neq i}|w^k|^2} dw^j
\end{aligned}$$

Substep 2.4: Compute $\bar{\partial}(\partial\phi_i)$.
We have $\partial\phi_i = \sum_{k\neq i} \frac{\bar{w}^k}{\rho} dw^k$. Now:

$$\begin{aligned}
\bar{\partial}(\partial\phi_i) &= \sum_{k\neq i} \bar{\partial}\left(\frac{\bar{w}^k}{\rho}\right) \wedge dw^k \\
&= \sum_{k\neq i} \left[\frac{\bar{\partial}(\bar{w}^k)}{\rho} - \frac{\bar{w}^k\bar{\partial}\rho}{\rho^2}\right] \wedge dw^k
\end{aligned}$$

Now, $\bar{\partial}(\bar{w}^k) = d\bar{w}^k$ and $\bar{\partial}\rho = \sum_{j\neq i} w^j d\bar{w}^j$:

$$\begin{aligned}
\bar{\partial}(\partial\phi_i) &= \sum_{k\neq i}\left[\frac{d\bar{w}^k}{\rho} - \frac{\bar{w}^k}{\rho^2}\sum_{j\neq i} w^j d\bar{w}^j\right] \wedge dw^k \\
&= \sum_{k\neq i}\frac{d\bar{w}^k \wedge dw^k}{\rho} - \sum_{k\neq i}\sum_{j\neq i}\frac{\bar{w}^k w^j}{\rho^2} d\bar{w}^j \wedge dw^k \\
&= \sum_{k\neq i}\frac{d\bar{w}^k \wedge dw^k}{\rho} - \sum_{j,k\neq i}\frac{\bar{w}^k w^j}{\rho^2} d\bar{w}^j \wedge dw^k
\end{aligned}$$

Therefore:

$$\partial\bar{\partial}\phi_i = \sum_{k\neq i}\frac{d\bar{w}^k \wedge dw^k}{\rho} - \sum_{j,k\neq i}\frac{\bar{w}^k w^j}{\rho^2} d\bar{w}^j \wedge dw^k$$

Step 3: Extract metric components in detail.
From the expression:

$$\partial\bar{\partial}\phi_i = \sum_{k\neq i}\frac{d\bar{w}^k \wedge dw^k}{\rho} - \sum_{j,k\neq i}\frac{\bar{w}^k w^j}{\rho^2} d\bar{w}^j \wedge dw^k$$

We can write this as:

$$\partial\bar{\partial}\phi_i = \sum_{j,k\neq i} h_{jk} dw^j \wedge d\bar{w}^k$$

where the metric components are:

$$h_{jk} = \frac{\partial^2\phi_i}{\partial w^j \partial\bar{w}^k}$$

Detailed calculation of h_{jk}:
We have $\phi_i = \log\rho$ where $\rho = 1 + \sum_{\ell\neq i} w^\ell \bar{w}^\ell$.
First, compute $\frac{\partial\phi_i}{\partial w^j}$:

$$\begin{aligned}
\frac{\partial\phi_i}{\partial w^j} &= \frac{\partial}{\partial w^j}(\log\rho) \\
&= \frac{1}{\rho}\frac{\partial\rho}{\partial w^j} \\
&= \frac{1}{\rho}\frac{\partial}{\partial w^j}\left(1 + \sum_{\ell\neq i} w^\ell \bar{w}^\ell\right) \\
&= \frac{1}{\rho}\bar{w}^j
\end{aligned}$$

Now, compute $\frac{\partial^2 \phi_i}{\partial \bar{w}^k \partial w^j}$:

$$\begin{aligned}
h_{jk} &= \frac{\partial}{\partial \bar{w}^k}\left(\frac{\bar{w}^j}{\rho}\right) \\
&= \frac{1}{\rho}\frac{\partial \bar{w}^j}{\partial \bar{w}^k} + \bar{w}^j \frac{\partial}{\partial \bar{w}^k}\left(\frac{1}{\rho}\right) \\
&= \frac{\delta_{jk}}{\rho} + \bar{w}^j \left(-\frac{1}{\rho^2}\right)\frac{\partial \rho}{\partial \bar{w}^k} \\
&= \frac{\delta_{jk}}{\rho} - \frac{\bar{w}^j}{\rho^2}\frac{\partial}{\partial \bar{w}^k}\left(1 + \sum_{\ell \neq i} w^\ell \bar{w}^\ell\right) \\
&= \frac{\delta_{jk}}{\rho} - \frac{\bar{w}^j}{\rho^2} w^k \\
&= \frac{\delta_{jk}}{\rho} - \frac{\bar{w}^j w^k}{\rho^2}
\end{aligned}$$

Factoring:

$$h_{jk} = \frac{1}{\rho}\left(\delta_{jk} - \frac{\bar{w}^j w^k}{\rho}\right) = \frac{\rho\delta_{jk} - \bar{w}^j w^k}{\rho^2}$$

Step 4: Verify the formula.
We have:

$$\omega|_{U_i} = \frac{\sqrt{-1}}{2}\sum_{j,k\neq i} h_{jk} dw^j \wedge d\bar{w}^k = \frac{\sqrt{-1}}{2}\partial\bar{\partial}\log \frac{\sum_{l=0}^n |z^l|^2}{|z^i|^2}$$

This metric is well-defined on $\mathbb{CP}^n$ and positive definite because the matrix (h_{jk}) is positive definite for all points in U_i. □

2 Chern Connection and Chern Curvature

Exercise 6.4

Problem: Show that the curvature of constant Hermitian metric on the tangent bundle of a complex torus $\mathbb{C}^n/\Gamma$ is trivial. Is this true for any Hermitian metric on the tangent bundle of $\mathbb{C}^n/\Gamma$?

Proof. **Part 1: Constant Hermitian metric.**

Let $\mathbb{C}^n/\Gamma$ be a complex torus where Γ is a lattice in $\mathbb{C}^n$. The tangent bundle $T(\mathbb{C}^n/\Gamma)$ can be trivialized using the standard coordinates $(z^1, \ldots, z^n)$ inherited from $\mathbb{C}^n$.

A constant Hermitian metric h on $T(\mathbb{C}^n/\Gamma)$ is given by:

$$h = \sum_{i,j=1}^n h_{ij} dz^i \otimes d\bar{z}^j$$

where h_{ij} are constant complex numbers with (h_{ij}) being a positive definite Hermitian matrix.

Since the metric is constant, in the trivialization given by the standard frame $\{\partial/\partial z^1, \ldots, \partial/\partial z^n\}$:

$$\frac{\partial h_{ij}}{\partial z^k} = 0 \quad \text{and} \quad \frac{\partial h_{ij}}{\partial \bar{z}^k} = 0$$

The Chern connection in this trivialization has connection 1-form:

$$\Gamma^i_{jk} = h^{i\bar{\ell}} \frac{\partial h_{\bar{\ell}k}}{\partial z^j} = 0$$

The curvature 2-form is:

$$\Theta^i_j = d\Gamma^i_j + \Gamma^i_k \wedge \Gamma^k_j$$

Since $\Gamma^i_j = 0$, we have:

$$\Theta^i_j = 0$$

Therefore, the curvature vanishes.

Part 2: General Hermitian metric.

For a non-constant Hermitian metric on $\mathbb{C}^n/\Gamma$, the curvature need not vanish.

Example: Consider the metric:

$$h = \sum_{i=1}^{n} f(z, \bar{z}) dz^i \otimes d\bar{z}^i$$

where f is a positive function that is periodic with respect to Γ but not constant.

For such a metric:

$$\frac{\partial h_{ij}}{\partial z^k} \neq 0 \quad \text{or} \quad \frac{\partial h_{ij}}{\partial \bar{z}^k} \neq 0$$

The connection forms Γ^i_j will be non-zero, and in general, the curvature:

$$\Theta^i_j = d\Gamma^i_j + \Gamma^i_k \wedge \Gamma^k_j \neq 0$$

Conclusion: The curvature vanishes for constant Hermitian metrics, but not necessarily for all Hermitian metrics on $\mathbb{C}^n/\Gamma$. □

Exercise 6.5

Problem: Let (E, h) be a Hermitian holomorphic vector bundle on a complex manifold X and ∇ be the Chern connection. For $s \in C^\infty(X, E)$, locally written as $s = s^\alpha e_\alpha$, prove that:

$$\nabla_{\frac{\partial}{\partial z^i}} \nabla_{\frac{\partial}{\partial z^j}} s^\beta - \nabla_{\frac{\partial}{\partial z^j}} \nabla_{\frac{\partial}{\partial z^i}} s^\beta = \Theta^\alpha_{\beta ij} s^\alpha$$

Proof. Let $\{e_\alpha\}$ be a local holomorphic frame for E and $s = s^\alpha e_\alpha$ be a smooth section.

Step 1: Apply the Chern connection.

The Chern connection acts on s as:

$$\nabla_{\frac{\partial}{\partial z^i}} s = \left(\frac{\partial s^\beta}{\partial z^i} + \Gamma^\beta_{\alpha i} s^\alpha \right) e_\beta$$

where $\Gamma^\beta_{\alpha i}$ are the connection coefficients (Christoffel symbols).

Step 2: Apply the connection twice.

First application:

$$\nabla_{\frac{\partial}{\partial z^i}} s = \frac{\partial s^\beta}{\partial z^i} e_\beta + s^\alpha \Gamma^\beta_{\alpha i} e_\beta$$

Second application on the β-component:

$$\begin{aligned}
\nabla_{\frac{\partial}{\partial z^j}} \nabla_{\frac{\partial}{\partial z^i}} s &= \nabla_{\frac{\partial}{\partial z^j}} \left[\left(\frac{\partial s^\beta}{\partial z^i} + \Gamma^\beta_{\alpha i} s^\alpha \right) e_\beta \right] \\
&= \frac{\partial}{\partial z^j} \left(\frac{\partial s^\beta}{\partial z^i} + \Gamma^\beta_{\alpha i} s^\alpha \right) e_\beta + \left(\frac{\partial s^\beta}{\partial z^i} + \Gamma^\beta_{\alpha i} s^\alpha \right) \Gamma^\gamma_{\beta j} e_\gamma
\end{aligned}$$

Expanding:

$$\begin{aligned}
\nabla_{\frac{\partial}{\partial z^j}} \nabla_{\frac{\partial}{\partial z^i}} s^\beta = \frac{\partial^2 s^\beta}{\partial z^j \partial z^i} + \frac{\partial \Gamma^\beta_{\alpha i}}{\partial z^j} s^\alpha + \Gamma^\beta_{\alpha i} \frac{\partial s^\alpha}{\partial z^j} \\
+ \Gamma^\gamma_{\beta j} \frac{\partial s^\gamma}{\partial z^i} + \Gamma^\gamma_{\beta j} \Gamma^\gamma_{\alpha i} s^\alpha
\end{aligned}$$

Step 3: Compute the commutator in detail.
Similarly, compute $\nabla_{\frac{\partial}{\partial z^i}} \nabla_{\frac{\partial}{\partial z^j}} s^\beta$:

$$\begin{aligned}
\nabla_{\frac{\partial}{\partial z^i}} \nabla_{\frac{\partial}{\partial z^j}} s^\beta &= \nabla_{\frac{\partial}{\partial z^i}} \left[\frac{\partial s^\beta}{\partial z^j} + \Gamma^\beta_{\alpha j} s^\alpha \right] \\
&= \frac{\partial}{\partial z^i} \left(\frac{\partial s^\beta}{\partial z^j} \right) + \frac{\partial}{\partial z^i} (\Gamma^\beta_{\alpha j} s^\alpha) + \Gamma^\gamma_{\beta i} \left(\frac{\partial s^\gamma}{\partial z^j} + \Gamma^\gamma_{\alpha j} s^\alpha \right) \\
&= \frac{\partial^2 s^\beta}{\partial z^i \partial z^j} + \frac{\partial \Gamma^\beta_{\alpha j}}{\partial z^i} s^\alpha + \Gamma^\beta_{\alpha j} \frac{\partial s^\alpha}{\partial z^i} \\
&\quad + \Gamma^\gamma_{\beta i} \frac{\partial s^\gamma}{\partial z^j} + \Gamma^\gamma_{\beta i} \Gamma^\gamma_{\alpha j} s^\alpha
\end{aligned}$$

Now compute the commutator term by term:

$$\begin{aligned}
&\nabla_{\frac{\partial}{\partial z^i}} \nabla_{\frac{\partial}{\partial z^j}} s^\beta - \nabla_{\frac{\partial}{\partial z^j}} \nabla_{\frac{\partial}{\partial z^i}} s^\beta \\
&= \left[\frac{\partial^2 s^\beta}{\partial z^i \partial z^j} + \frac{\partial \Gamma^\beta_{\alpha j}}{\partial z^i} s^\alpha + \Gamma^\beta_{\alpha j} \frac{\partial s^\alpha}{\partial z^i} + \Gamma^\gamma_{\beta i} \frac{\partial s^\gamma}{\partial z^j} + \Gamma^\gamma_{\beta i} \Gamma^\gamma_{\alpha j} s^\alpha \right] \\
&\quad - \left[\frac{\partial^2 s^\beta}{\partial z^j \partial z^i} + \frac{\partial \Gamma^\beta_{\alpha i}}{\partial z^j} s^\alpha + \Gamma^\beta_{\alpha i} \frac{\partial s^\alpha}{\partial z^j} + \Gamma^\gamma_{\beta j} \frac{\partial s^\gamma}{\partial z^i} + \Gamma^\gamma_{\beta j} \Gamma^\gamma_{\alpha i} s^\alpha \right]
\end{aligned}$$

Term-by-term cancellation:

- $\frac{\partial^2 s^\beta}{\partial z^i \partial z^j} - \frac{\partial^2 s^\beta}{\partial z^j \partial z^i} = 0$ (equality of mixed partials)
- The derivative terms give:
$$\left(\frac{\partial \Gamma^\beta_{\alpha j}}{\partial z^i} - \frac{\partial \Gamma^\beta_{\alpha i}}{\partial z^j} \right) s^\alpha$$
- For the $\frac{\partial s^\alpha}{\partial z^i}$ and $\frac{\partial s^\gamma}{\partial z^j}$ terms, relabel indices ($\alpha \leftrightarrow \gamma$ in second expression):
$$\Gamma^\beta_{\alpha j} \frac{\partial s^\alpha}{\partial z^i} + \Gamma^\gamma_{\beta i} \frac{\partial s^\gamma}{\partial z^j} - \Gamma^\beta_{\alpha i} \frac{\partial s^\alpha}{\partial z^j} - \Gamma^\gamma_{\beta j} \frac{\partial s^\gamma}{\partial z^i}$$

Renaming $\gamma \to \alpha$:

$$\Gamma^{\beta}_{\alpha j}\frac{\partial s^{\alpha}}{\partial z^{i}} + \Gamma^{\alpha}_{\beta i}\frac{\partial s^{\alpha}}{\partial z^{j}} - \Gamma^{\beta}_{\alpha i}\frac{\partial s^{\alpha}}{\partial z^{j}} - \Gamma^{\alpha}_{\beta j}\frac{\partial s^{\alpha}}{\partial z^{i}} = 0$$

These terms cancel by rearranging.

- The quadratic connection terms give:

$$\left(\Gamma^{\gamma}_{\beta i}\Gamma^{\gamma}_{\alpha j} - \Gamma^{\gamma}_{\beta j}\Gamma^{\gamma}_{\alpha i}\right) s^{\alpha}$$

After cancellation, we get:

$$\begin{aligned}&\nabla_{\frac{\partial}{\partial z^{i}}}\nabla_{\frac{\partial}{\partial z^{j}}} s^{\beta} - \nabla_{\frac{\partial}{\partial z^{j}}}\nabla_{\frac{\partial}{\partial z^{i}}} s^{\beta}\\ &= \left(\frac{\partial \Gamma^{\beta}_{\alpha i}}{\partial z^{j}} - \frac{\partial \Gamma^{\beta}_{\alpha j}}{\partial z^{i}} + \Gamma^{\gamma}_{\beta j}\Gamma^{\gamma}_{\alpha i} - \Gamma^{\gamma}_{\beta i}\Gamma^{\gamma}_{\alpha j}\right) s^{\alpha}\end{aligned}$$

Step 4: Identify with curvature.
By definition, the curvature tensor is:

$$\Theta^{\alpha}_{\beta ij} = \frac{\partial \Gamma^{\beta}_{\alpha i}}{\partial z^{j}} - \frac{\partial \Gamma^{\beta}_{\alpha j}}{\partial z^{i}} + \Gamma^{\beta}_{\gamma i}\Gamma^{\gamma}_{\alpha j} - \Gamma^{\beta}_{\gamma j}\Gamma^{\gamma}_{\alpha i}$$

Note the index position. With proper index arrangement:

$$\nabla_{\frac{\partial}{\partial z^{i}}}\nabla_{\frac{\partial}{\partial z^{j}}} s^{\beta} - \nabla_{\frac{\partial}{\partial z^{j}}}\nabla_{\frac{\partial}{\partial z^{i}}} s^{\beta} = \Theta^{\alpha}_{\beta ij} s^{\alpha}$$

This completes the proof. □

Exercise 6.6

Problem: Let (E, h) be a Hermitian holomorphic vector bundle of rank r over a complex n-manifold X with Chern connection ∇.

1. If (E, h) is Nakano positive or dual Nakano positive, then (E, h) is Griffiths positive.
2. (E, h) is Nakano positive if and only if (E^*, h^*) is dual Nakano negative.

Proof. **Part (1): Nakano positive implies Griffiths positive.**
Recall the definitions:

- *Griffiths positive:* For any non-zero $v \in T_xX$ and $s \in E_x$:

$$\sum_{i,j,\alpha,\beta} \Theta_{\beta\bar{\alpha}ij} s^{\alpha}\bar{s}^{\beta} v^{i}\bar{v}^{j} > 0$$

- *Nakano positive:* For any non-zero sections $s \in E$ and $v \in TX$:

$$\sum_{i,j,\alpha,\beta} \Theta_{\beta\bar{\alpha}ij} s^{\alpha}\bar{s}^{\beta} v^{i}\bar{v}^{j} > 0$$

when evaluated pointwise.

Actually, Nakano positivity is stronger than Griffiths positivity. To show Griffiths positivity, we need to show that for any decomposable tensor $v \otimes s$, the curvature form is positive.

For Nakano positivity, consider the Hermitian form:

$$Q(v \otimes s) = \sum_{i,j,\alpha,\beta} \Theta_{\beta\bar{\alpha}ij} s^\alpha \bar{s}^\beta v^i \bar{v}^j$$

This is a Hermitian form on $TX \otimes E$. Nakano positivity says this form is positive definite on all of $TX \otimes E$.

Griffiths positivity only requires positivity on decomposable tensors $v \otimes s$. Since every decomposable tensor is in $TX \otimes E$, and Nakano positivity implies $Q > 0$ on all vectors, it certainly implies $Q > 0$ on decomposable tensors.

Therefore, Nakano positive $\Rightarrow$ Griffiths positive.

For dual Nakano positivity, the argument is similar but applied to E^* instead of E.

Part (2): Nakano positive iff (E^*, h^*) is dual Nakano negative.

Let Θ be the curvature of (E, h) and Θ^* be the curvature of (E^*, h^*).

The relationship between these curvatures is:

$$(\Theta^*)_{\alpha\bar{\beta}ij} = -\Theta_{\beta\bar{\alpha}ij}$$

This follows from the fact that the Chern connection on E^* is the dual connection, and curvatures transform with a sign change.

Now, (E, h) is Nakano positive means:

$$\sum_{i,j,\alpha,\beta} \Theta_{\beta\bar{\alpha}ij} s^\alpha \bar{s}^\beta v^i \bar{v}^j > 0$$

for all non-zero $s \in E$ and $v \in TX$.

For (E^*, h^*), let $\sigma \in E^*$ be a non-zero section. Dual Nakano negativity means:

$$\sum_{i,j,\alpha,\beta} (\Theta^*)_{\beta\bar{\alpha}ij} \sigma^\alpha \bar{\sigma}^\beta v^i \bar{v}^j < 0$$

Substituting the relationship $(\Theta^*)_{\alpha\bar{\beta}ij} = -\Theta_{\beta\bar{\alpha}ij}$:

$$\sum_{i,j,\alpha,\beta} (-\Theta_{\beta\bar{\alpha}ij}) \sigma^\alpha \bar{\sigma}^\beta v^i \bar{v}^j < 0$$

This is equivalent to:

$$\sum_{i,j,\alpha,\beta} \Theta_{\beta\bar{\alpha}ij} \sigma^\alpha \bar{\sigma}^\beta v^i \bar{v}^j > 0$$

which is exactly the Nakano positivity condition for (E, h) (considering σ as representing elements in E).

Therefore, (E, h) is Nakano positive if and only if (E^*, h^*) is dual Nakano negative. $\square$

3 Chern Class

Exercise 6.7

Problem: Prove the following formula by definition:

$$c_1(E \otimes F) = \mathrm{rk}(F) \cdot c_1(E) + \mathrm{rk}(E) \cdot c_1(F)$$

where E, F are complex vector bundles on a complex manifold.

Proof. The first Chern class can be computed using the curvature of a connection.

Step 1: Recall the definition.

For a complex vector bundle E with connection ∇^E and curvature Θ^E, the first Chern class is:

$$c_1(E) = \frac{\sqrt{-1}}{2\pi} \mathrm{Tr}(\Theta^E)$$

Step 2: Connection on tensor product.

For $E \otimes F$ with connections ∇^E and ∇^F, the induced connection is:

$$\nabla^{E\otimes F}(s \otimes t) = (\nabla^E s) \otimes t + s \otimes (\nabla^F t)$$

Step 3: Curvature of tensor product.

The curvature of $E \otimes F$ can be computed using the formula:

$$\Theta^{E\otimes F} = \Theta^E \otimes \mathrm{Id}_F + \mathrm{Id}_E \otimes \Theta^F$$

This means that in a local trivialization where E has rank r and F has rank s, the curvature $\Theta^{E\otimes F}$ is an $(rs) \times (rs)$ matrix that can be written in block form.

Step 4: Compute the trace in detail.

Let $\{e_1, \dots, e_r\}$ be a local holomorphic frame for E (rank r) and $\{f_1, \dots, f_s\}$ be a local holomorphic frame for F (rank s). Then:

$$\{e_i \otimes f_j : i = 1, \dots, r;\ j = 1, \dots, s\}$$

is a local holomorphic frame for $E \otimes F$ (rank rs).

Index notation: Label the basis elements of $E\otimes F$ by pairs (i, j) where $i \in \{1, \dots, r\}$ and $j \in \{1, \dots, s\}$.

The curvature matrix has components $(\Theta^{E\otimes F})_{(i,j),(k,\ell)}$ acting on the basis vector $e_k \otimes f_\ell$ to produce a linear combination.

From the formula $\Theta^{E\otimes F} = \Theta^E \otimes \mathrm{Id}_F + \mathrm{Id}_E \otimes \Theta^F$:

$$\begin{aligned}
\Theta^{E\otimes F}(e_k \otimes f_\ell) &= (\Theta^E \otimes \mathrm{Id}_F)(e_k \otimes f_\ell) + (\mathrm{Id}_E \otimes \Theta^F)(e_k \otimes f_\ell) \\
&= \Theta^E(e_k) \otimes f_\ell + e_k \otimes \Theta^F(f_\ell) \\
&= \sum_{i=1}^r (\Theta^E)_{ik} e_i \otimes f_\ell + \sum_{j=1}^s e_k \otimes (\Theta^F)_{j\ell} f_j \\
&= \sum_{i=1}^r (\Theta^E)_{ik} (e_i \otimes f_\ell) + \sum_{j=1}^s (\Theta^F)_{j\ell} (e_k \otimes f_j)
\end{aligned}$$

Therefore:

$$(\Theta^{E\otimes F})_{(i,j),(k,\ell)} = (\Theta^E)_{ik}\delta_{j\ell} + \delta_{ik}(\Theta^F)_{j\ell}$$

The trace is the sum of diagonal elements:

$$
\begin{aligned}
\mathrm{Tr}(\Theta^{E\otimes F}) &= \sum_{i=1}^{r}\sum_{j=1}^{s}(\Theta^{E\otimes F})_{(i,j),(i,j)} \\
&= \sum_{i=1}^{r}\sum_{j=1}^{s}\left[(\Theta^{E})_{ii}\delta_{jj} + \delta_{ii}(\Theta^{F})_{jj}\right] \\
&= \sum_{i=1}^{r}\sum_{j=1}^{s}\left[(\Theta^{E})_{ii}\cdot 1 + 1\cdot(\Theta^{F})_{jj}\right] \\
&= \sum_{i=1}^{r}\sum_{j=1}^{s}(\Theta^{E})_{ii} + \sum_{i=1}^{r}\sum_{j=1}^{s}(\Theta^{F})_{jj} \\
&= \sum_{j=1}^{s}\sum_{i=1}^{r}(\Theta^{E})_{ii} + \sum_{i=1}^{r}\sum_{j=1}^{s}(\Theta^{F})_{jj} \\
&= s\cdot\sum_{i=1}^{r}(\Theta^{E})_{ii} + r\cdot\sum_{j=1}^{s}(\Theta^{F})_{jj} \\
&= s\cdot\mathrm{Tr}(\Theta^{E}) + r\cdot\mathrm{Tr}(\Theta^{F})
\end{aligned}
$$

where $r = \mathrm{rk}(E)$ and $s = \mathrm{rk}(F)$.
Step 5: Apply the definition of c_1.
Therefore:

$$
\begin{aligned}
c_1(E\otimes F) &= \frac{\sqrt{-1}}{2\pi}\mathrm{Tr}(\Theta^{E\otimes F}) \\
&= \frac{\sqrt{-1}}{2\pi}[s\cdot\mathrm{Tr}(\Theta^{E}) + r\cdot\mathrm{Tr}(\Theta^{F})] \\
&= s\cdot\frac{\sqrt{-1}}{2\pi}\mathrm{Tr}(\Theta^{E}) + r\cdot\frac{\sqrt{-1}}{2\pi}\mathrm{Tr}(\Theta^{F}) \\
&= \mathrm{rk}(F)\cdot c_1(E) + \mathrm{rk}(E)\cdot c_1(F)
\end{aligned}
$$

This completes the proof. □

Exercise 6.8

Problem: Compute

$$\int_{\mathbb{CP}^n} c_1(T\mathbb{CP}^n)^n$$

by the Fubini-Study metric on $\mathbb{CP}^n$.

Proof. **Step 1: Compute the first Chern class.**
The tangent bundle of $\mathbb{CP}^n$ fits into the Euler sequence:

$$0 \to \mathbb{C} \to \mathcal{O}(1)^{\oplus(n+1)} \to T\mathbb{CP}^n \to 0$$

From this sequence, we get:

$$c(T\mathbb{CP}^n) = \frac{c(\mathcal{O}(1)^{\oplus(n+1)})}{c(\mathbb{C})} = (1+h)^{n+1}$$

where $h = c_1(\mathcal{O}(1))$ is the hyperplane class.
Expanding:

$$c(T\mathbb{CP}^n) = 1 + (n+1)h + \binom{n+1}{2}h^2 + \cdots + h^{n+1}$$

Therefore:

$$c_1(T\mathbb{CP}^n) = (n+1)h$$

Step 2: Compute $c_1(T\mathbb{CP}^n)^n$.
Using the cup product:

$$\begin{aligned} c_1(T\mathbb{CP}^n)^n &= [(n+1)h]^n \\ &= (n+1)^n h^n \end{aligned}$$

Step 3: Evaluate the integral in detail.
We need to compute:

$$\int_{\mathbb{CP}^n} h^n$$

The hyperplane class $h = c_1(\mathcal{O}(1))$ is represented by the Fubini-Study form. On the chart U_0 with coordinates $w^i = z^i/z^0$ for $i = 1, \ldots, n$:

$$\omega = \frac{\sqrt{-1}}{2}\partial\bar{\partial}\log(1 + |w|^2)$$

where $|w|^2 = \sum_{j=1}^n |w^j|^2$.

For normalization, we have $h = [\omega]$ in cohomology (often with factor 2π).

Explicit calculation:
Let $\rho = 1 + \sum_{j=1}^n |w^j|^2$. From Exercise 6.3, we computed:

$$\omega = \frac{\sqrt{-1}}{2}\sum_{j,k=1}^n h_{jk} dw^j \wedge d\bar{w}^k$$

where:

$$h_{jk} = \frac{\delta_{jk}}{\rho} - \frac{\bar{w}^j w^k}{\rho^2}$$

The n-th power of ω is:

$$\omega^n = \left(\frac{\sqrt{-1}}{2}\right)^n n! \det(h_{jk}) dw^1 \wedge d\bar{w}^1 \wedge \cdots \wedge dw^n \wedge d\bar{w}^n$$

Compute $\det(h_{jk})$:
Using the matrix determinant lemma, for $h_{jk} = \frac{1}{\rho}(\delta_{jk} - \frac{\bar{w}^j w^k}{\rho})$:

$$\begin{aligned} \det(h_{jk}) &= \det\left(\frac{1}{\rho}\right)^n \det\left(\delta_{jk} - \frac{\bar{w}^j w^k}{\rho}\right) \\ &= \frac{1}{\rho^n} \cdot \frac{1}{1 + \frac{|w|^2}{\rho}} \quad \text{(using rank-1 perturbation formula)} \\ &= \frac{1}{\rho^n} \cdot \frac{\rho}{\rho + |w|^2} \\ &= \frac{1}{\rho^n} \cdot \frac{\rho}{\rho + |w|^2} \\ &= \frac{1}{\rho^{n+1}} \end{aligned}$$

Therefore:

$$\omega^n = \left(\frac{\sqrt{-1}}{2}\right)^n \frac{n!}{\rho^{n+1}} dw^1 \wedge d\bar{w}^1 \wedge \cdots \wedge dw^n \wedge d\bar{w}^n$$

Integrate over $\mathbb{CP}^n$:
Using $\mathbb{CP}^n = U_0 \cup \dots$ (covering), on $U_0 \cong \mathbb{C}^n$:

$$\int_{\mathbb{CP}^n} \omega^n = \int_{\mathbb{C}^n} \left(\frac{\sqrt{-1}}{2}\right)^n \frac{n!}{(1+|w|^2)^{n+1}} dw^1 \wedge d\bar{w}^1 \wedge \cdots \wedge dw^n \wedge d\bar{w}^n$$

Using $dw^j \wedge d\bar{w}^j = -2i dx^j \wedge dy^j$ where $w^j = x^j + iy^j$:

$$\int_{\mathbb{C}^n} \frac{dw^1 \wedge d\bar{w}^1 \wedge \cdots \wedge dw^n \wedge d\bar{w}^n}{(1+|w|^2)^{n+1}} = (2i)^n \int_{\mathbb{R}^{2n}} \frac{dx^1 dy^1 \cdots dx^n dy^n}{(1+|w|^2)^{n+1}}$$

In polar coordinates (r, θ) where $|w|^2 = r^2$, the measure is $r^{2n-1} dr \cdot d\Omega_{2n-1}$:

$$\int_{\mathbb{R}^{2n}} \frac{dx^1 dy^1 \cdots dx^n dy^n}{(1+r^2)^{n+1}} = \mathrm{Vol}(S^{2n-1}) \int_0^\infty \frac{r^{2n-1}}{(1+r^2)^{n+1}} dr$$

Using $\mathrm{Vol}(S^{2n-1}) = \frac{2\pi^n}{(n-1)!}$ and the integral:

$$\int_0^\infty \frac{r^{2n-1}}{(1+r^2)^{n+1}} dr = \frac{(n-1)!}{2 \cdot n!}$$

We get:

$$\int_{\mathbb{CP}^n} \omega^n = \frac{\pi^n}{n!}$$

Normalization: With standard normalization $h = [2\pi\omega]$ (or setting $\int_{\mathbb{CP}^n} h^n = 1$), we have:

$$\int_{\mathbb{CP}^n} h^n = 1$$

Step 4: Final answer.
Therefore:

$$\int_{\mathbb{CP}^n} c_1(T\mathbb{CP}^n)^n = (n+1)^n \int_{\mathbb{CP}^n} h^n = (n+1)^n \cdot 1 = \boxed{(n+1)^n}$$

$\square$